

Lecture 8 (10/2)

Correlation (2.4), Estimators, Bias, and Consistency

Standardized variables

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Consider the standardized variables

$$Z_X = (X - \mu_X)/\sigma_X, \quad Z_Y = (Y - \mu_Y)/\sigma_Y$$

What is the regression coefficient of Z_Y on Z_X ?

Standardized variables

$$Z_Y = (Y - \mu_Y) / \sigma_Y$$

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Therefore

$$\boxed{\beta_1 \sigma_X / \sigma_Y = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}}$$

This is called the correlation coefficient and denoted ρ .

$1 - \rho^2$ as a ratio of variances

$$\sigma_Y^2 = \beta_1^2 \sigma_X^2 + \sigma_\epsilon^2 \quad (\text{variance of } Y = \beta_0 + \beta_1 X + \epsilon)$$

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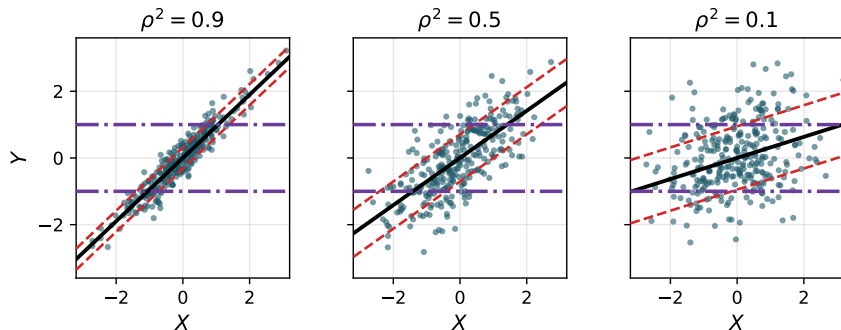
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$1 - \rho^2$ is the fraction of the variance in Y *left over* after conditioning on X .

High and low ρ^2



Every panel has $\mu_X = \mu_Y = 0$ and $\sigma_X = \sigma_Y = 1$. Black: $E[Y | X] = \rho X$. Red: $\pm \sigma_\epsilon = \pm \sqrt{1 - \rho^2}$. Purple: $\pm \sigma_Y = \pm 1$.

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- $\hat{\theta}$ is random – it will vary depending on which sample we collect.
- θ is a fixed, non-random number (in frequentist statistics).

Sample distribution and standard error

Example

Let $Y_1, \dots, Y_N \sim \text{Normal}(\mu, \sigma^2)$ and estimate μ with $\hat{\mu} = \bar{Y} = \frac{1}{N} \sum_i Y_i$.

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$$\hat{\mu} \sim \text{Normal}\left(\mu, \frac{\sigma^2}{N}\right).$$

This is the sample distribution of $\hat{\mu}$: its distribution over replications of the data. Its standard deviation is the standard error

$$\text{se}(\hat{\mu}) = \frac{\sigma}{\sqrt{N}}.$$

Estimating q

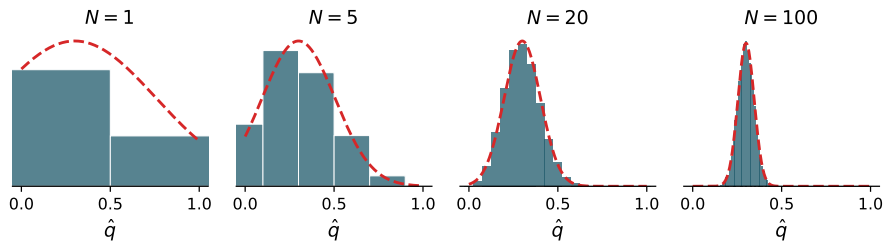
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Let $Y_1, \dots, Y_N \sim \text{Bernoulli}(q)$ and estimate q with $\hat{q} = \overline{Y}$.

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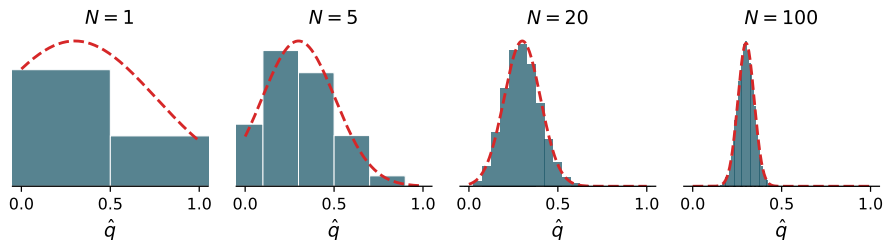
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$\hat{q}$ is not Normal, but by the CLT the Normal result carries over for large N , with $\sigma^2 = q(1 - q)$:

$$\hat{q} \approx \text{Normal} \left(q, \frac{q(1 - q)}{N} \right), \quad \text{se}(\hat{q}) = \sqrt{\frac{q(1 - q)}{N}}.$$

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$\hat{\theta}$ is consistent if

$$E[\hat{\theta}] \rightarrow \theta \quad \text{and} \quad \text{se}(\hat{\theta}) \rightarrow 0 \quad \text{as } N \rightarrow \infty.$$

Example

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For $Y_i \sim \text{Normal}(\mu, \sigma^2)$, is $\hat{\mu}_2 = \bar{Y} + 1/2$ biased? Is it consistent? Find its sample distribution.

Example

$$\hat{\sigma}_0^2 = \frac{1}{n} \sum_i (Y_i - \bar{Y})^2$$

Is this unbiased?

The bias-variance decomposition

$$\text{MSE}_{\hat{\theta}} = E[(\hat{\theta} - \theta)^2] = \text{var}(\hat{\theta}) + (E[\hat{\theta}] - \theta)^2.$$

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Error from an estimator splits into how *variable* it is, plus how far off it is *on average*.